NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

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TOPIC: Project Monitoring and control



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CONCEPTS COVERED

- Earned value analysis
- Baseline budget



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Earned value analysis

- Earned value analysis has gained in popularity in recent years and may be seen as a refinement of the cost monitoring.
- It originated in the USA's Department of Defence (DOD) as a part of a set of measures to control projects being carried out by contractors for the DOD.
- It is based on assigning a 'value' to each task or work package based on the original expenditure forecasts.





Earned value analysis cont ...

- One way of looking at this is as the equivalent of the price that might be agreed by a contractor to do some unit of work.
- The assigned value is the original budgeted cost for the item and is known as the *planned value* (PV) or *budgeted cost of work scheduled* (BCWS).
- A task that has not started is assigned an earned value of zero and when it has been completed, it, and hence the project, is credited with the original planned value of the task.





Earned value analysis cont ...

- The total value credited to a project at any point is known as the *earned value*(EV) or *budgeted cost of work performed* (BCWP) and this can be represented as a money value, an amount of staff time or as a percentage of the PV.
- EV is thus analogous to the agreed price to be paid to the contractor once the work is completed.

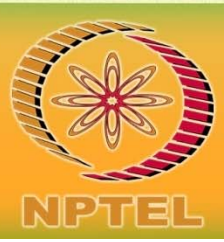




Earned value analysis cont ...

- Where tasks have been started but are not yet complete, some consistent method of assigning an earned value must be applied.
- Common methods in software projects are
  - *The 0/100 technique*
  - *The 50/50 technique*
  - *The 75/25 technique*
  - *The milestone technique*
  - *The percentage complete technique*





Earned value analysis cont ...

- The 0/100 technique is preferred for software development.
- The 50/50 technique can give a false sense of security by over-valuing the reporting of activity starts.
- The milestone technique might be appropriate for activities with a long duration estimate but, in such cases, it is better to break that activity into a number of smaller ones.





The 0/100 technique

- A task is assigned a value of zero until such time that it is completed, when it is given a value of 100% of the budgeted value.

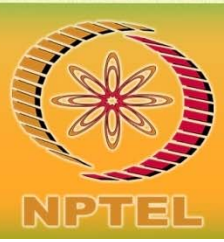




The 50/50 technique

- A task is assigned a value of 50% of its value as soon as it is started and then given a value of 100% once it is complete
  - This matches some contractual arrangements where a contractor is given half the agreed price when starting the work, perhaps to help pay for raw materials, and the remainder on successful completion.





The 75/25 technique

- The task is assigned 75% on starting and 25% on completion
  - This is often used when a large item of equipment is being bought:
    - 75% is paid when the equipment is actually delivered and the rest when installation and testing have been satisfactorily completed.





The milestone technique

- A task is given a value based on the achievement of milestones that have been assigned values as part of the original budget plan.

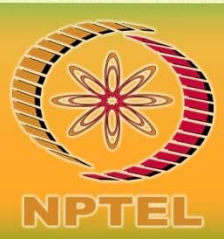




The percentage complete

- In some cases there may be a way of objectively measuring the amount of work completed
  - For example, as part of the implementation of an information system, a number of data records have to be manually typed into a database.
  - The actual number so far completed can be objectively counted and the value can be assigned accordingly.





Earned value analysis

- *Planned value (PV)* or *Budgeted cost of work scheduled (BCWS)* – original estimate of the effort/cost to complete a task (compare with idea of a ‘price’)
- *Earned value (EV)* or *Budgeted cost of work performed* (BCWP) – total of PVs for the work completed at this time



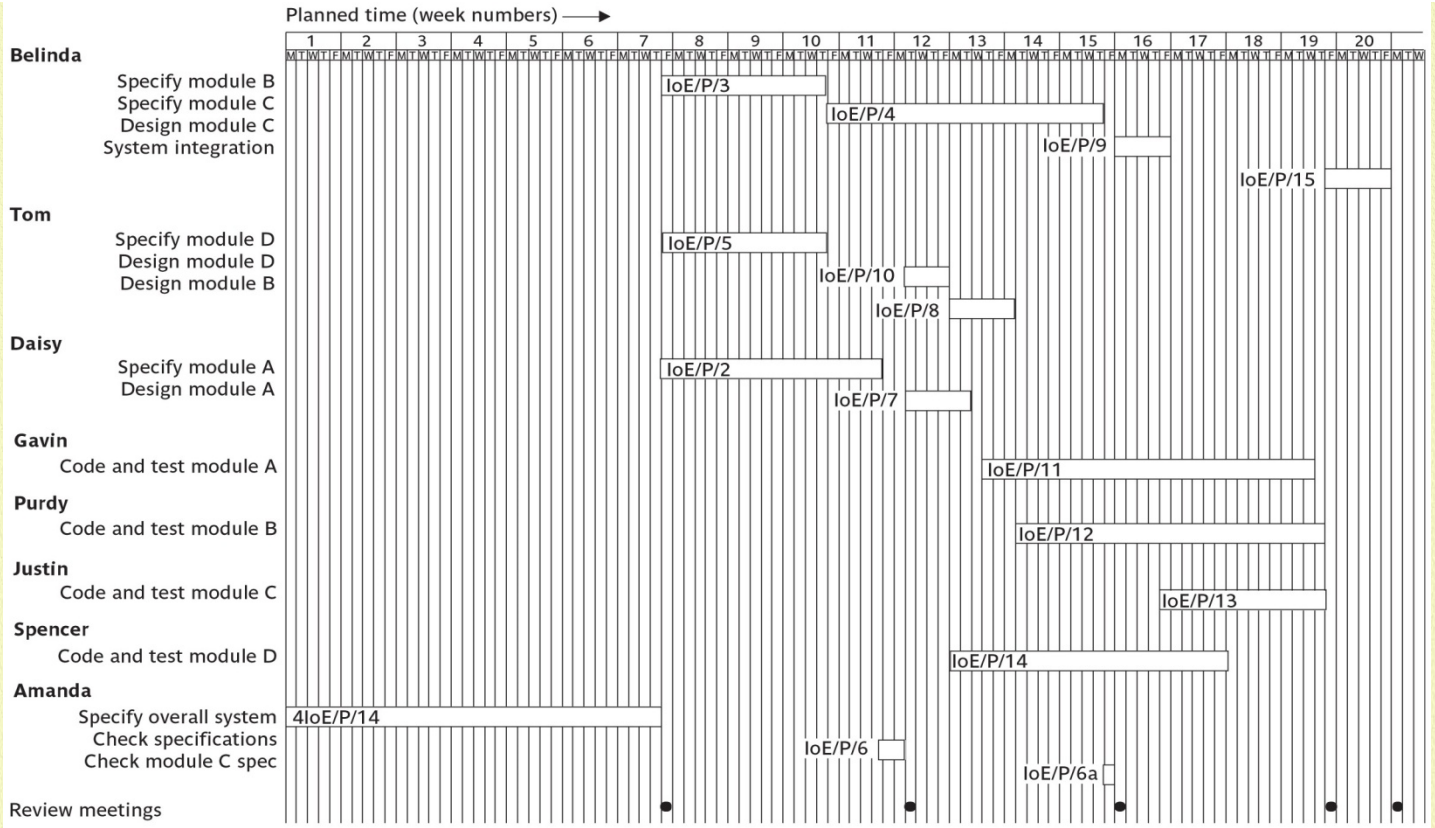


The baseline budget

- The first step in setting up an earned value analysis is to create the *baseline budget*.
- The *baseline budget* is based on the project plan and shows the forecast growth in earned value through time.
- Earned value may be measured in monetary values but, in the case of staff-intensive projects such as software development, it is common to measure earned value in person-hours and workdays.







A Sample Gantt chart

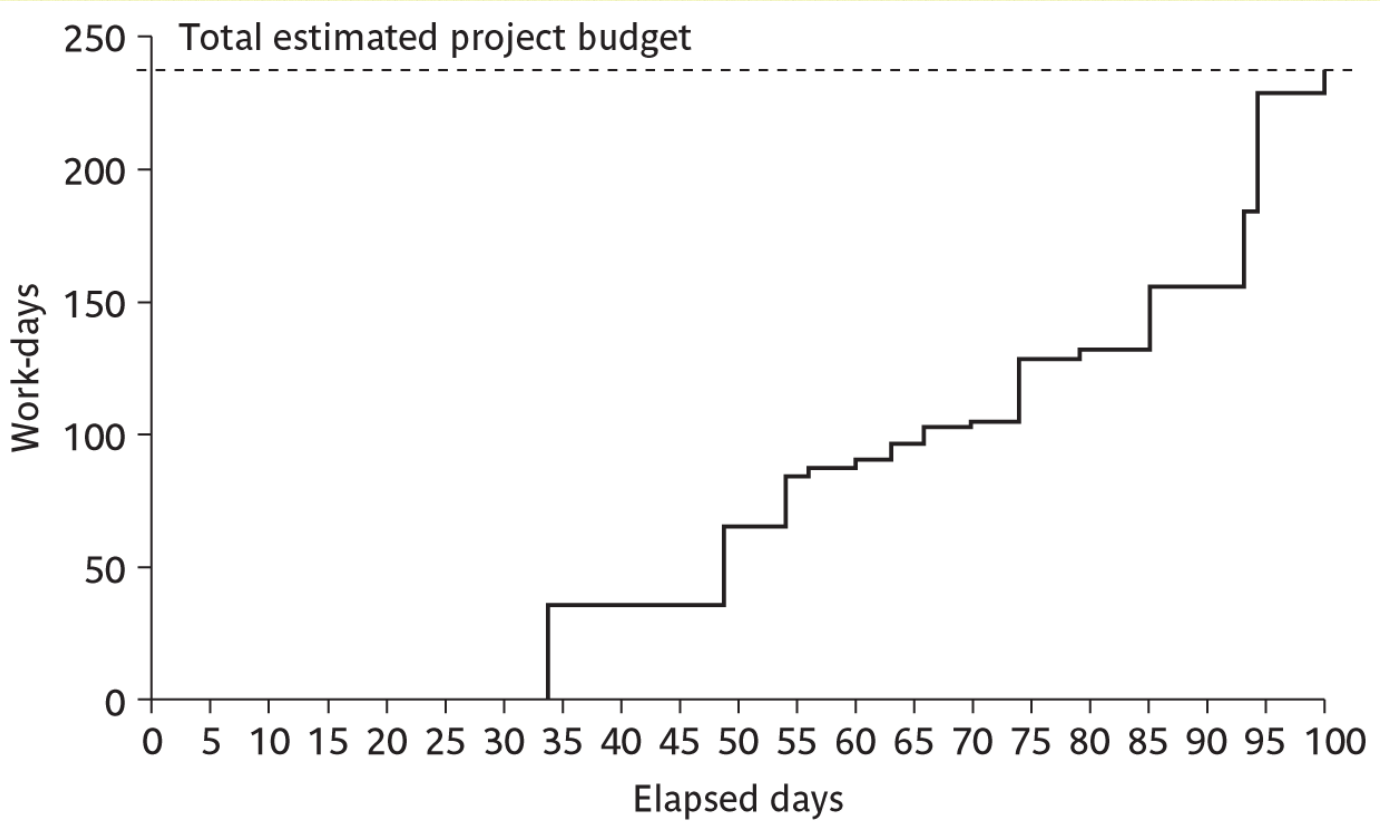
Example baseline budget calculation

Task	Budgeted workdays	Scheduled completion	Cumulative workdays	% cumulative earned value
Specify overall system	34	34	34	14.35
Specify module B	15	49	64	27.00
Specify module D	15	49		
Specify module A	20	54	84	35.44
Check specifications	2	56	86	36.28
Design module D	4	60	90	37.97
Design module A	7	63	97	40.93
Design module B	6	66	103	43.46
Specify module C	25	74	128	54.01
Check module C spec	1	75	129	54.43
Design module C	4	79	133	56.12
Code and test module D	25	85	158	66.67
Code and test module A	30	93	188	79.32
Code and test module B	28	94	231	97.47
Code and test module C	15	94		
System integration	6	100	237	100.00





The baseline budget



The baseline budget cont ...

- The example baseline budget uses 0/100 technique for crediting earned value to the project.
- This project is not expected to be credited with any earned value until day 34, when the activity ‘specify overall system’ is to be completed.
- This activity was forecast to consume 34 person-days and it will therefore be credited with 34 person-days of earned value when it has been completed.
- The other steps in the baseline budget chart coincide with the scheduled completion dates of other activities.





Summary

- Discussed briefly earned value analysis
  - T*he 0/100 technique*
  - *The 50/50 technique*
  - *The 75/25 technique*
  - *The milestone technique*
  - *The percentage complete technique*
- Presented the concept baseline budget





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.






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